A standards-based knowledge graph that bridges scientific workflows, run-time provenance, and tool registries

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Abstract

Life science workflows are now prevalent for implementing, executing, and sharing complex data analyses, increasing their scalability and reproducibility. Adhering to the FAIR principles for software further reinforces their reproducibility and the reliability of their results. To maximize their FAIRness, consistent and standardised annotations are critical across several levels: workflows, individual steps, software tools, and input/output data. Such comprehensive metadata make workflows easier to understand, reuse and reproduce, while keeping track of the provenance of their results. However, a unified, queryable knowledge framework that integrates workflows with enriched metadata is lacking. To address this, we developed an integrated workflow knowledge base, that consolidates FAIR metadata from diverse sources and workflow languages into a standardised graph-based representation. It leverages established ontologies and standards (e.g. EDAM, schema.org) to enrich metadata, and link the workflow structure with its execution traces. Our approach provides FAIR-compliant metadata of publicly available pipelines, enabling queries at every granularity level, while accounting for the quality of source data annotation.

Keywords

Knowledge graphs, Ontologies, FAIR, Workflows, SPARQL,

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1. Introduction

The rapid expansion of high-throughput technologies in the life sciences has generated an unprecedented amount of data. While this wealth of information opens new avenues for biological discovery, it also creates a set of practical problems. Existing analytical methods and software often cannot cope with the volume and complexity of the data, and the resulting pipelines become difficult to reproduce [1]. Poor reproducibility undermines transparency and hampers the reuse or extension of previous analyses, which in turn slows scientific progress.

To address these issues, the community has turned to scientific workflows to formally capture the steps of the data-analysis process. Workflows provide automation, scalability, and, crucially, reproducibility [2]. The uptake of this approach has resulted in a very large number of publicly available, open-source workflows [3, 4]. More broadly, such workflows are part of a large landscape of digital objects that also includes the datasets they consume and generate, as well as the software required to execute the analysis steps.

Because workflows are systematic descriptions of analysis pipelines, they are natural candidates for the FAIR principles (Findable, Accessible, Interoperable, Reusable) [5, 6], which were originally formulated for data [7] and later extended to software [8]. By making workflows FAIR, we can improve their quality, traceability, and the reliability of the results they produce. Applying FAIR to workflows therefore requires comprehensive, standardised metadata that describes their structure, execution environment, and provenance. Standardised annotations—based on established ontologies, vocabularies, schemas, and formats—allow workflows to be queried as a knowledge base, to expose usage rights and software requirements, to be interpreted across different platforms, and to be understood at multiple levels of granularity. Such a knowledge base would also support seamless integration, discovery, and reuse of workflow components [5].

Extracting metadata from the many heterogeneous workflow formats commonly used in Life Sciences, as well as linked objects such as their execution traces or software components, is therefore a central challenge to feed such knowledge bases. Our contribution lies in automatically harvesting the essential FAIR metadata and normalising it in a single, graph-based representation.

The resulting knowledge graph, leveraging state-of-the-art and community adopted ontologies/controlled vocabularies inter-links and enriches a corpus of workflow-centric metadata: it connects the structural description of a workflow to its execution traces, software dependencies, and semantic annotations, and makes all of these connections queryable through a unified knowledge model. In other words, the knowledge graph adds a semantic layer on top of existing registries, turning a collection of files into an interoperable, searchable knowledge base.

For users, this brings several concrete advantages:

- **Researchers** can locate every workflow that implements a specific analytical step (e.g., "variant calling with GATK") across all supported languages, compare implementations, and reuse the most appropriate one.
- **Tool developers** obtain a consolidated view of which software packages are already wrapped in workflows, what versions are used, and where gaps exist, guiding the creation of new wrappers.
- **Data-managers and curators** can assess the FAIR compliance of a workflow at a glance, automatically generate citation and licensing information, and integrate workflows into larger pipelines.

In this work we introduce a *knowledge graph* that represents, stores, and shares standardised descriptions of publicly available workflows, authored in a variety of languages and frameworks, and linking their structural descriptions to execution traces and the software they run, by leveraging existing standards. This graph can be used by a software module we also introduce, to assess the quality of software functional annotations.

The paper is organised as follows. First, we review the standards selected for tool description, workflow representation, and the underlying graph model. Next, we describe the construction of the knowledge graph and demonstrate how it enables queries at any level of workflow detail. We then present a toolbox for retrieving and assessing the informativeness tools' semantic annotations. Finally, we summarise our contributions and outline directions for future research.

2. Scientific workflow and tool metadata standards

In order to fully describe a workflow, it is essential to select an appropriate and detailed representation model. This model should account for all levels of workflow granularity that can be examined at multiple scales (see Figure 1). At the broadest level, workflows are considered as single objects that process and analyse input datasets to generate results within a particular scientific context. At a finer granularity, these workflows comprise multiple steps, either workflows themselves (sub-workflows) or single atomic operations. At the most specific level, these processes integrate and invoke the specific tools, software and scripts required to fulfil the task.

To comprehensively understand and represent the complexity of workflows, each of these precision levels should be considered. They should also be evaluated from two distinct provenance perspectives: a prospective provenance, which describes their structure and the way they are interconnected, and a retrospective provenance that captures the execution traces of workflows, subworkflows, processes and software, as well as the actual value of their parameters, inputs, and outputs at runtime.

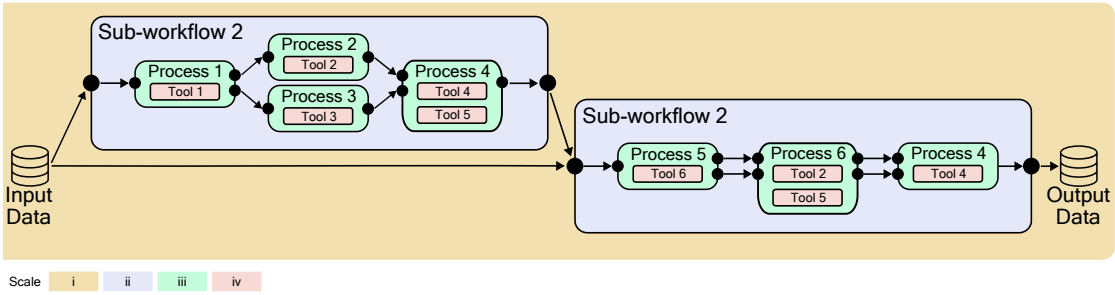


Figure 1: Representation of workflow annotation granularity levels: scale i: Workflow, ii: Sub-workflows, iii: Processes, iv: Tools

2.1. Workflow specification and provenance standards

We have reviewed and investigated several annotation standards, with the requirement that they support the representation of workflows at every granularity level (Table 1). This includes details on their scientific domain, general metadata, structure, and execution traces. Ontologies extending the W3C PROVenance Ontology[1] (PROV-O) for provenance description, such as P-PLAN [9], D-PROV [10] and PROV-One [11] support the representation of workflow plans and their executions. The provenance framework Common Workflow Language Prov (CWLProv) [12] also builds on PROV-O, providing a domain-independent, open-source, system-neutral, and community-driven approach for standardising provenance metadata and enabling workflow re-execution. PROV-O has also been instrumental to increase the findability and reuse of bioinformatics data thanks to domain ontologies such as EDAM [13]. While PROV-based approaches support fine-grained and logically sound provenance descriptions, more recent approaches, based on the widely adopted Schema.org vocabulary, have attracted significant attention across many communities.

One such approach, the Provenance Run RO-Crate profile (Provenance Run Crate) [14] of the RO-Crate standard[2], emerges as a robust solution for representing comprehensive pipelines. Tailored for workflow description, it reflects their multi-level granularity while covering their metadata, prospective and retrospective provenance. Provenance Run Crate includes two other, less detailed profiles: (i) Process Run Crate characterises software used for file creation, (ii) Workflow Run Crate combines Process Run Crate and WorkflowHub's [4] Workflow RO-Crate profile[3] for the representation of a workflow, its datasets and metadata. Workflow Run Crate describes the execution of a workflow by a Workflow Management System (WMS), including its inputs and outputs, but not its individual steps nor the software used.

[1]https://www.w3.org/TR/prov-o/
[2]https://www.researchobject.org/ro-crate/
[3]https://about.workflowhub.eu/Workflow-RO-Crate/

These profiles are interoperable with several WMSs, and workflow repositories. Their adoption by the WMS Galaxy [15], the Nextflow [16] plugin nf-prov[4] and the WorkflowHub [4] repository facilitates the extraction of annotations while ensuring their compatibility and ease of integration with existing resources. Furthermore, RO-Crate uses schema.org[5] and Bioschemas[6], two widely recognised and extensively utilised schemas, thereby enhancing its interoperability with other standards.

Scope \ Granularity	Workflows	Process	Software	Data
Domain	Context (*about*): OBI [17], SIO, EFO, EDAM topics, SO [18], GO [19, 20], NCIT [21], STATO[7], CHEBI [22], ISSVA [23, 24], CVDO [25], NIFSTD [26]			
	Operations (*potentialAction*): EDAM operation, OBI	Operations: EDAM operation, OBI	Repositories (bio.tools, Bioconda [27], Biocontainers [28], DebianMed[8] [29]) Operations: OBI, EDAM operation	Data type (*additionalType*): OBI, EDAM data, NIDM results [30], SIO
Metadata	RO-Crate with Bioschema and Schema.org			
			CodeMeta[9], SWO [31]	BIDS [32], NWB [33], Format (*encodingFormat*): EDAM format, schema.org
Prospective and retrospective provenance	Provenance Run Crate, PROV-O, P-PLAN			

Table 1
Recommended standards for workflow annotation at various granularity levels. *about*, *potentialAction*, *additionalType* and *encodingFormat* correspond to schema.org properties.

2.2. Domain-specific standards

As RO-Crate is based on Linked Data principles[10] and is written in JSON-LD format, it is modular and extendable. Thus, it can integrate ontologies or vocabularies from external sources to complement the Provenance Run Crate's workflow descriptions. EDAM [34, 35] is an example of such a standard. This biomedical ontology is made of four sub-ontologies: EDAM Topic, EDAM Operation, EDAM Data and EDAM Format. These sub-ontologies cover the bioinformatics domain, operations, data types and data formats, respectively. These standards describe the workflows, their processes and their data. However, they fall short in providing comprehensive annotation at the finest level: the software used at each steps. In that domain, bio.tools [36, 37]

[4]https://github.com/nextflow-io/nf-prov
[5]https://schema.org/
[6]http://bioschemas.org/
[10]http://w3.org/DesignIssues/LinkedData.html

provides a solution to uniquely identify scientific software, and annotate them using EDAM. bio.tools is a widely used repository encompassing more than 30,000 tools covering a large spectrum of bioinformatics domain. In bio.tools, each tool is annotated with EDAM topics, operations, and type and format of their input and output data.

3. Tools and workflow knowledge graph

3.1. Workflow representation model

Using the previously selected standards, we have then developed a comprehensive workflow representation model (Figure 2) capable of representing the multiple granularity levels needed, from the individual tools to the entire workflows.

The model relies heavily on Provenance Run Crate and encompasses Workflow Run Crate and WorkflowHub's workflow profiles. Notably, it also includes an extension of the latter, that decouples workflow entities from their script files [38]. We have enriched these profiles with further information on processes, by specifying their inputs/outputs and their order with the schema.org properties *nextItem*[11] and *previousItem*[12]. Additionally, we have chosen existing schema.org properties to integrate other standards, thereby extending workflow annotations with consistent terminologies (Table 1). For example, the property *about*[13] introduces domain ontologies like EDAM Topic, Experimental Factor Ontology (EFO) [39] and Semanticscience Integrated Ontology (SIO) [40]. They describe workflow context, steps, and data. The properties *encodingFormat*[14] and *additionalType*[15] refer to standards like EDAM Data or EDAM Format, for data description. By integrating and combining these data and standards, we ensure consistency and standardisation, enabling robust querying and retrieving of information.

3.2. ShareFAIR-KG knowledge base

Once the representation model defined, we built ShareFAIR-KG[16], the knowledge base, made of the following components: (i) The workflows extracted from the main workflow repositories (WorkflowHub, Galaxy IWC), and corresponding to the main WMSs (Nextflow, CWL, Galaxy, see Figure 3). (ii) To query this knowledge base, a SPARQL[17] endpoint was deployed into a Jena Fuseki[18] server, inside a dedicated Docker[19] container. SPARQL is suitable for this application,

[11] https://schema.org/nextItem
[12] https://schema.org/previousItem
[13] https://schema.org/about
[14] https://schema.org/encodingFormat
[15] https://schema.org/additionalType
[16] https://gitlab.liris.cnrs.fr/sharefair/knowledge_base_workflow_annotations/ShareFAIR-KG
[17] https://www.w3.org/TR/sparql11-overview/
[18] https://jena.apache.org/documentation/fuseki2/index.html
[19] https://www.docker.com/

since it is dedicated to query RDF[20] graphs supporting schema evolution through integration of new data sources and relationships. It also adheres to the Linked Data Principles, effectively capturing relationships, semantic and hierarchy between data entities, which is particularly relevant for ontologies. (iii) Domain specific ontologies like EDAM, EFO and SIO, as well as the content of WorkflowHub and of bio.tools repositories, ensuring unique software and annotation referencing and annotation. (iv) Finally, newly generated RO-Crate metadata describing workflow structure and execution traces.

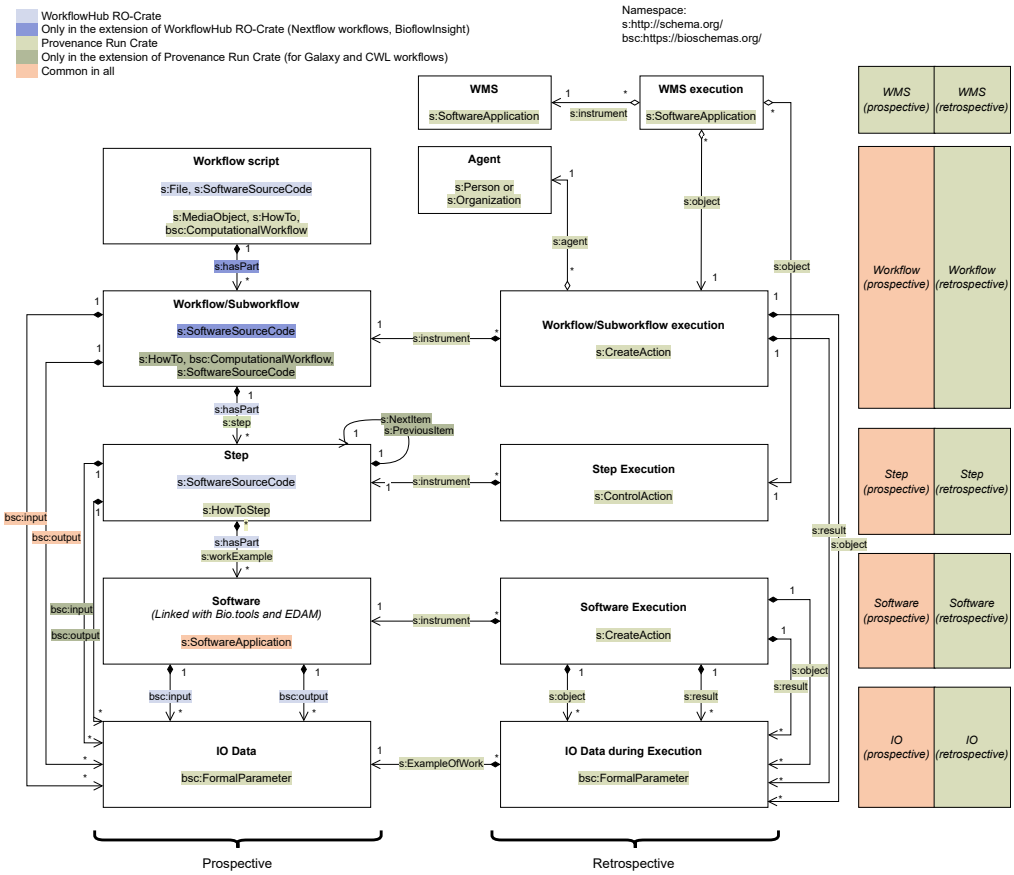


Figure 2: Workflow representation model. The workflow granularity levels are detailed on the right for the prospective and the retrospective provenance. The colors indicate which properties are used in which RO-Crate profiles: the profile used in WorkflowHub and its extension (used in RO-Crate metadata created with BioflowInsight for Nextflow workflow) ; and Provenance Run RO-Crate and its extension, used for CWL and Galaxy workflows.

[20] https://www.w3.org/RDF/

Workflows from three different WMS were selected: Common Workflow Language (CWL) [41, 42], Nextflow and Galaxy. They were extracted from WorkflowHub, well-known by the community for promoting workflow FAIRness using open tools and standards like bio.tools and EDAM. Some of the Nextflow workflows are part of the nf-core[21] framework, while all galaxy workflows were taken from the *Intergalactic Workflow Commission* (IWC)[22] library. Both provide open-source pipelines following best practices. IWC workflows are versioned, reviewed and tested. Since they include test data, they facilitate their re-execution and the recording of their execution traces. WorkflowHub and Galaxy server come with the added benefit of using or generating RO-Crate for workflow description, following respectively WorkflowHub's workflow profile and Workflow Run Crate profile. Their RO-Crate metadata files can therefore serve as a foundation for automatically generating more detailed workflow annotations.

Workflows ultimately extracted and integrated into the ShareFAIR-KG adhere to certain guidelines or specifications to ensure their compliance with the chosen representation model:

- Their repository is available for cloning.
- The code of CWL and Nextflow workflows is parsable and analysable with our scripts.
- Test files of IWC Galaxy workflow are provided and they are re-executable on the server usegalaxy.org using the command-line tool Planemo [43].
- All workflows possess a valid license. We selected workflows with open-source and permissive license compatible with our requirements[23].

The newly produced RO-Crate metadata files were generated differently depending on the WMS. First, Nextflow workflows metadata were generated using BioflowInsight [44]. These RO-Crates follow an extension of WorkflowHub's workflow profile [38] and they detail their prospective provenance. Then, the code of CWL workflows was parsed, to extract their prospective provenance and generate corresponding Provenance Run Crate metadata files. Finally, Galaxy workflow repositories were cloned and their tests were executed on usegalaxy.org server[24], using the command-line tool Planemo which creates output files detailing execution traces. Workflow Run Crate metadata files generated by Galaxy were exported. The Planemo output files and Workflow Run Crate metadata files were combined, to produce a complete Provenance Run Crate record. When possible, a link was established between the tools executed in Galaxy and their bio.tools identifiers. Thus, metadata providing a description of the structure of Nextflow and CWL workflows, and a description of both the structure and the execution traces of Galaxy pipelines, was generated in compliance with the chosen representation model.

[21] https://nf-co.re/
[22] https://iwc.galaxyproject.org/
[23] Workflows with licenses CC-BY-NC-ND-4.0 and CC-BY-SA-4.0 were not integrated into the knowledge base upon concerns of rights to use in this context.
[24] https://usegalaxy.org/

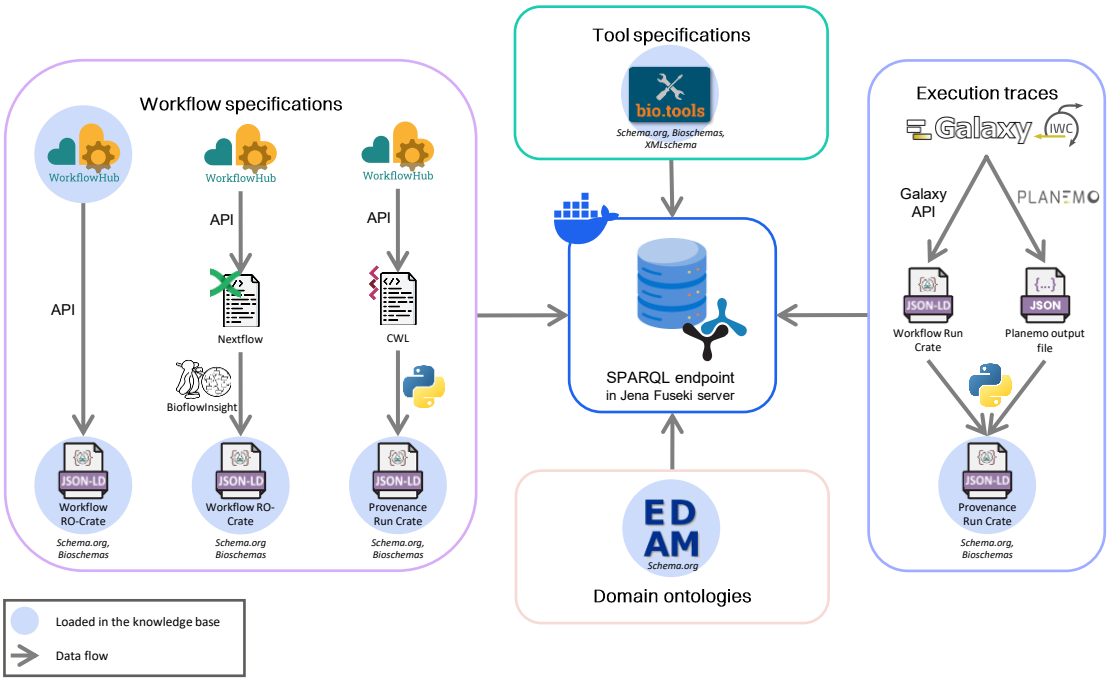


Figure 3: Sources of the workflow metadata integrated in the knowledge base. Blue circles indicate the files loaded in the knowledge base, with the schema they use in italic. The knowledge base is a SPARQL endpoint deployed with Jena Fuseki server in a Docker container. It is populated with tool specifications from bio.tools and domain ontologies like EDAM. RO-Crate metadata files are also loaded to provide information on the structure of workflows from WorkflowHub. They either follow WorkflowHub's RO-Crate profile for workflows (Nextflow workflows) or Provenance Run RO-Crate profile (CWL workflows). Execution traces are generated for IWC Galaxy workflows in the form of Provenance Run Crate metadata files.

4. Knowledge base querying

Once populated, the knowledge graph contains prospective provenance information for 49 Nextflow workflows and 73 CWL workflows, and Provenance Run Crate metadata on the structure and execution traces of 102 Galaxy workflows (Table 2). Each Galaxy workflow might have multiple executions. Once integrated in the knowledge graph, these workflows can be queried using SPARQL queries.

In the next paragraphs, we present four use cases that illustrate the practical applications of our knowledge base across different scenarios.

Tool adoption. The first use case identifies the most widely used tools from bio.tools, along with their associated operations, to analyze software adoption in existing workflows. The cor-

Table 2
Counts of the most represented entities in the Knowledge Graph, unifying Nextflow and CWL workflows from WorkflowHub, Galaxy workflows from IWC, and tools metadata from bio.tools (via the Research Sofware Ecosystem [45] RDF dump).

Entity	Instances
Workflows (total)	875
Workflows with only WorkflowHub's metadata	651
Workflows with additional metadata	224
Nextflow workflows	49
Galaxy workflows	102
CWL workflows	73
Workflow steps	2,177
Workflow inputs	3,520
Workflow outputs	1,749
Workflow executions	492
Step executions	2,625
Input values	284
Output values	105
Tools (source: bio.tools)	30,328
EDAM classes	3,522

responding query is shown in Figure 4, with key findings summarized in Table 3. Results show that MultiQC [46] (dedicated to quality control report generation) is the most used tool in our workflow base, followed by dada2 [47] (dedicated to sample sequence inference from amplicon data). With citation counts exceeding 9,000 and 32,000 respectively, it is unsurprising that these tools are widely integrated into numerous workflows. EDAM annotations specify the operations performed by those tools (Sequence quality control, DNA barcoding and Variant calling), providing meaningful insights on the use of software and workflows in the knowledge base.

```
SELECT ?biotools_url ?biotools_name ?edam_operation (COUNT(?biotools_url) AS ?
      usageCount)
WHERE {
    ?workflow_id a bsc:ComputationalWorkflow ;
        schema:name ?workflow_name ;
        schema:step ?step .
    ?step a schema:HowToStep;
        schema:workExample ?tool .
    ?tool a schema:SoftwareApplication ;
        schema:identifier ?biotools_url .
    ?biotools_url schema:propertyID "biotools identifier" .
    ?biotools_url a schema:SoftwareApplication ;
        schema:name ?biotools_name ;
        schema:featureList ?biotools_features .
    ?biotools_features rdfs:label ?edam_operation .
}
GROUP BY ?biotools_url ?biotools_name ?edam_operation
ORDER BY DESC(?usageCount)
LIMIT 10
```

Figure 4: SPARQL query to identify the most used bio.tools tools and their operations.

Table 3
Results of the SPARQL query retrieving most-used bio.tools tools, and their integration frequency in steps (# workflow steps). The names of the tools are from bio.tools. EDAM operations indicates the label of the EDAM term qualifying the operation performed by the tool in bio.tools. The namespace edam corresponds to http://edamontology.org/.

Tool name	bio.tools identifier	EDAM operation	EDAM operation label	# workflow steps
MultiQC	https://bio.tools/multiqc	edam:operation_3218, edam:operation_2428	Sequencing quality control, Validation	18
dada2	https://bio.tools/dada2	edam:operation_3200, edam:operation_3227	DNA barcoding, Variant calling	12
Bandage	https://bio.tools/bandage	edam:operation_3184	Sequence assembly visualisation	5
MACS	https://bio.tools/macs	edam:operation_3501, edam:operation_1781, edam:operation_3222	Enrichment analysis, Gene regulatory network analysis, Peak calling	5
QUAST	https://bio.tools/quast	edam:operation_3180, edam:operation_0337	Sequence assembly validation, Visualisation	5

Workflows using specific tools. In the second use case, we want to identify workflows that use a given tool (e.g. MultiQC, as a follow-up of the first use-case). Figure 4 shows the query to extract the workflows that use the software MultiQC, as defined in bio.tools [25]. Five of the resulting workflows are detailed in Table 4, along some metadata, such as their name, source code, license, and WMS.

```
PREFIX bsc: <https://bioschemas.org/>
PREFIX schema: <http://schema.org/>

SELECT DISTINCT ?workflow_id ?workflow_name ?source ?programming_language ?
      license
WHERE {
    ?workflow_dataset a schema:Dataset ;
        schema:hasPart ?workflow_id ;
        schema:isBasedOn ?source ;
        schema:license ?license .
    ?workflow_id a bsc:ComputationalWorkflow ;
        schema:name ?workflow_name ;
        schema:step ?step ;
        schema:programmingLanguage ?programming_language .
    ?step a schema:HowToStep;
        schema:workExample ?tool .
    ?tool a schema:SoftwareApplication ;
        schema:identifier ?biotools_url .
    ?biotools_url a schema:SoftwareApplication ;
        schema:name "MultiQC" .
    FILTER(STRSTARTS(STR(?biotools_url), "https://bio.tools/"))
}
LIMIT 5
```

Figure 5: SPARQL query to retrieve the name, the source url, the programming language and the license of 5 of the workflows which steps use the tool labelled "MultiQC" in bio.tools.

[25] https://bio.tools/multiqc

Table 4
Results of the SPARQL query retrieving five workflows using the tool MultiQC, with some of their metadata (source, programming language and license).

Workflow name	Workflow source	License	WMS
Genome Assembly from Hifi reads - VGP3	https://github.com/iwc-workflows/Assembly-Hifi-only-VGP3	CC-BY-4.0	galaxy
Scaffolding with Hi-C data VGP8	https://github.com/iwc-workflows/Scaffolding-HiC-VGP8	CC-BY-4.0	galaxy
ATACseq	https://github.com/iwc-workflows/atacseq	MIT	galaxy
ChIP-seq Analysis: Paired-End Read Processing	https://github.com/iwc-workflows/chipseq-pe	MIT	galaxy
ChIP-seq Analysis: Single-End Read Processing	https://github.com/iwc-workflows/chipseq-sr	MIT	galaxy

Workflow input formats. Beyond tool usage, the third use case investigates the input and output data of a workflow. Here, we want to extract the formats, as defined in EDAM, of the input data of a given workflow (e.g. gatk4w[26]). The SPARQL query that extracts this information is given in Figure 4. The resulting formats, detailed in Table 5, are mainly FASTA and FASTQ file formats.

```
PREFIX bsc: <https://bioschemas.org/>
PREFIX schema: <http://schema.org/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?format_label ?format ?input_name
WHERE {
    ?workflow a bsc:ComputationalWorkflow ;
        schema:name ?workflow_name ;
        <https://bioschemas.org/ComputationalWorkflow#input> ?wf_input .
    ?wf_input schema:encodingFormat ?format ;
        schema:name ?input_name .
    ?format rdfs:label ?format_label .
    FILTER(STR(?workflow_name) = STR("gatk4W"))
}
```

Figure 6: SPARQL query fetching EDAM format annotations of annotated inputs, for the CWL workflow gatk4w.

Table 5
Results of the SPARQL query fetching EDAM format of annotated inputs of the CWL workflow gatk4W.

Input name	EDAM format label	EDAM format
sars_cov_2_reference_genome	FASTA	http://edamontology.org/format_1929
rnaseq_left_reads	FASTQ	http://edamontology.org/format_1930
rnaseq_right_reads	FASTQ	http://edamontology.org/format_1930

Workflow tools output formats. In this fourth use case, information about workflow output data formats is extracted from bio.tools, which contains EDAM annotation on the type and format of tool outputs. More specifically, the EDAM format of some outputs of the Galaxy

[26]https://workflowhub.eu/workflows/30/

workflow ATACseq [48] can be retrieved using bio.tools identifiers. The SPARQL query is shown in Figure 4, and the extracted output formats are listed in Table 6. The main formats found in this workflow are BED, WIG, TSV, HTML, JSON, YAML, and CSV. These results stem from EDAM annotations in bio.tools, relying on the quality and the completeness of tool annotations.

```
PREFIX bsc: <https://bioschemas.org/>
PREFIX schema: <http://schema.org/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?workflow ?biotools_out ?format_label
WHERE {
    ?workflow a bsc:ComputationalWorkflow ;
        schema:name ?workflow_name ;
        schema:step ?step .
    ?step a schema:HowToStep;
        schema:workExample ?tool .
    ?tool a schema:SoftwareApplication ;
        schema:identifier ?tool_id ;
        schema:identifier ?tool_identifier .
    ?tool_identifier a schema:SoftwareApplication ;
        bsc:output ?biotools_out .
    ?biotools_out schema:encodingFormat ?format .
    ?format rdfs:label ?format_label .
    ?tool_id a schema:PropertyValue .
    FILTER(STR(?workflow_name) = STR("ATACseq"))
    FILTER(STRSTARTS(STR(?tool_identifier), STR(biotools:)))
}
```

Figure 7: SPARQL query fetching the EDAM format of annotated outputs of the Galaxy workflow ATACseq.

Table 6
Results of the SPARQL query retrieving EDAM format of annotated outputs of the Galaxy workflow ATACseq.

EDAM format label	EDAM format
BED	http://edamontology.org/format_3003
WIG	http://edamontology.org/format_3005
TSV	http://edamontology.org/format_3475
HTML	http://edamontology.org/format_2331
JSON	http://edamontology.org/format_3464
YAML	http://edamontology.org/format_3750
CSV	http://edamontology.org/format_3752

5. Enriching the knowledge graph with reasoning and assessing semantic annotation informativeness

Software and workflow repositories such as bio.tools or WorkflowHub already use EDAM as a terminology to expose metadata and facilitate search with bioinformatics reference terms. However, when registering tools or workflows, choosing the most relevant EDAM terms can be time-consuming and requires domain expertise. In addition, annotators are often not familiar with the reasoning capabilities encoded into an ontology. This leads to an important variability of tools and workflow annotations, depending on experts availability and expertise [49]. In

this section we propose a toolbox, for the knowledge graph, aimed at better exploiting the domain-specific reasoning capabilities provided by EDAM to identify the informativeness of semantically annotated tools and workflows.

The proposed ShareFAIR-KG[27] for tools and workflows makes it possible to query many aspects of the workflows, and especially the specific annotation of tools used in workflow steps. Among these tool annotations, the EDAM annotations as provided by bio.tools constitute an important source of information on the tools' topics, operations, data, and format.

The quality of this information relies on both its correctness (is the association between a tool and an annotation true?), and its informativeness (is the annotation precise or vague?). Assessing correctness is inherently difficult, as it requires comparison with a known true reference annotation dataset, which we generally do not have access to. Even without considering spurious annotations, the informativeness of a tool's description can be affected by three aspects: incompleteness, vagueness and obsolescence (due either to the evolution of the tools, or the evolution of the EDAM ontology itself). Coupling the knowledge graph with the EDAM hierarchy yields benefits on all three and makes it possible to quantify the information conveyed by a tool's annotations.

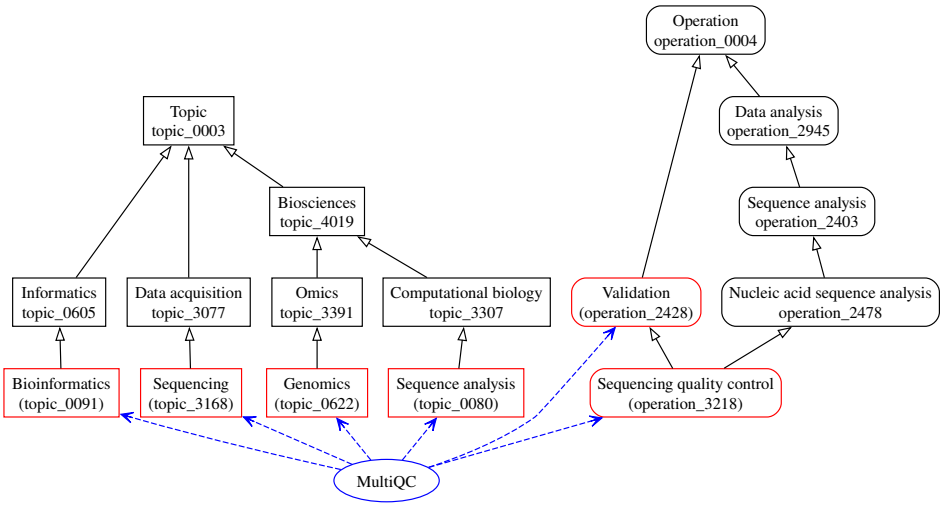


Figure 8: Direct (in red) and inferred (in black) topics and operations EDAM annotations for the MultiQC tool obtained with the `describe-viz` command from the EDAMAnnot toolbox. Dotted blue arrows indicate direct annotations, black plain arrows represent the EDAM hierarchy of annotations. Square boxes represent topics, and rounded boxes represent operations. The EDAM class hierarchy reveals annotations implicitly associated to a tool.

As an example, following the same principle as in table 3, the tool MultiQC has six direct topic and operation annotations, but when leveraging the EDAM class hierarchy, a total of sixteen annotations are retrieved as shown in Figure 8. This provides an exhaustive list of both the explicit direct annotations and implicit annotations which are inferred.

[27] https://gitlab.liris.cnrs.fr/sharefair/knowledge_base_workflow_annotations/ShareFAIR-KG

Also note that this allows to spot occasional redundancies among some direct annotations: the direct operation *Validation* is implied by the narrower class *Sequencing quality control*.

Among the annotations of a tool (direct as well as inferred), some are more precise than others. This can be estimated by different metrics depending on the task at hand. The simpler consists in counting the number of tools annotated by an annotation or one of its descendants. This results in highlighting general annotations that annotate many tools. Conversely, Information Content (the opposite of logarithm of the proportion of tools annotated by an annotation or one of its descendants) highlights annotations that annotate few tools. These are the most informative, but they can make it difficult to recognise that two tools having different precise annotations are indeed similar if the annotations are similar. Shannon entropy (the product of the information content of an annotation and the proportion of tools annotated by this annotation) offers a compromise by being low for the most general annotations (they have a small information content), and for the most precise (they annotate a small proportion of tools). Figure 9 illustrates how simple count, information content and Shannon entropy provide different perspectives on the annotations of the MultiQC tool.

5.1. EDAMannot toolbox

To facilitate these analyses, we developed a command-line Python toolbox named EDAMannot[28] using the knowledge graph introduced in section 3 as a data source.

It provides a command-line access and supports queries leveraging the EDAM hierarchy. The toolbox offers processing, metrics and visualisation features.

The first feature consists in describing the annotations of a given tool. The `describe` command takes one or several tool names and returns their direct and inherited annotations. The `describe-viz` alternative command generates a graph representing these annotations, colors them according to their informativeness, and highlights intersections between multiple tools.

The second feature focuses on annotation quality metrics through the `QC` command. This command reports annotation counts, Information Content (IC), and Shannon entropy.

5.2. Enriching knowledge graph with EDAM reasoning

5.2.1. Enrichment based on subClassOf hierarchy

Based on the bio.tools dataset described in the previous section, the EDAMannot toolbox identifies 264 unique topics (99,159 topic annotations) and 529 unique operations (69,106 operation annotations) for the 30,328 tools registered. Among all tools, 2,159 (7.12%) have no topic or operation annotations. On average, tools have 3.27 ± 1.46 (standard deviation) directly assigned topics and 2.28 ± 1.44 directly assigned operations. By considering inherited annotations computed by EDAMannot, this increases to 10.68 ± 5.03 topics and 8.26 ± 5.14 operations.

[28] https://github.com/ulysseLeclanche/EDAMannot

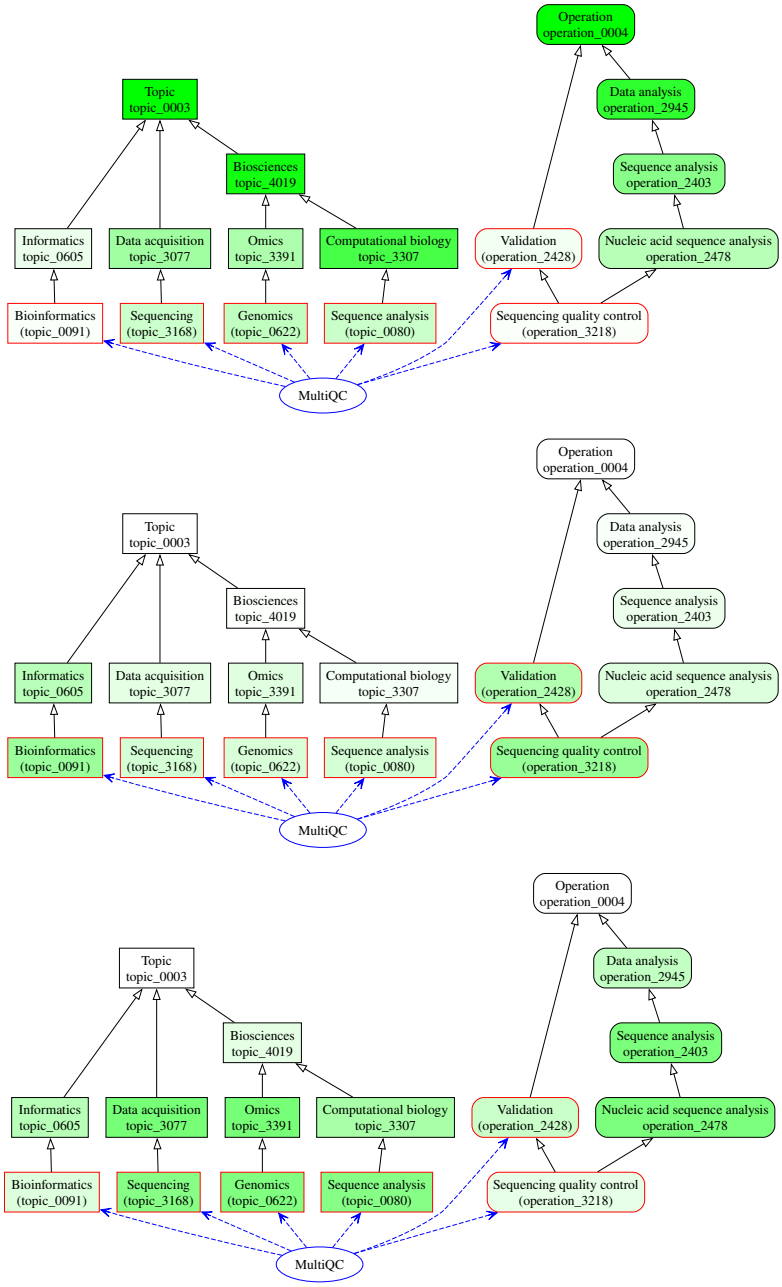
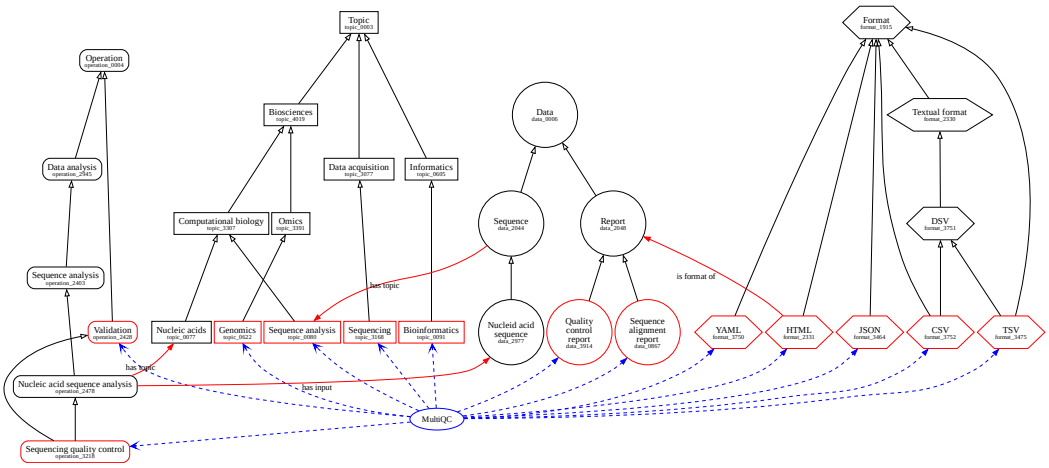


Figure 9: Three versions of annotations informativeness using the same conventions as Figure 8. Colors saturation indicates number of tools annotated (top), Information Content (middle), and Shannon entropy (bottom). For example for topics, tool count highlights general annotations such as *Topic* (`topic_003`) and *Biosciences* (`topic_4019`), Information Content highlights rare annotations such as *Bioinformatics* (`topic_0091`) and Shannon entropy highlights more balanced annotations such as *Sequencing* (`topic_3168`), *Genomics* (`topic_0622`) and *Sequence analysis* (`topic_0080`) but downplays the importance of *Bioinformatics*.

In addition, EDAMannot reveals annotation redundancies - i.e. direct annotations that are also ancestor annotations, therefore more general than a direct annotation - in 3,787 tools (12.45%) for topics and in 2,060 tools (6.79%) for operations.

Finally, EDAMannot highlights 1,119 obsolete annotations, including 190 obsolete topic annotations, 265 for operations, 546 for data, and 118 for formats. We identified 159 tools with at least 1 deprecated topic of the 8 deprecated topics annotating at least one tool. For operations, 1,051 tools containing at least one obsolete operation among the 20 obsolete operations annotating at least one tool were identified. Note that the EDAM ontology recommends 837 candidate annotations for the deprecated ones (and 5 of these 837 have since become deprecated themselves).

5.2.2. Enrichment based on inferred dependencies between data, topics, operations and formats



The EDAM ontology is not only defining a taxonomy of bioinformatics concepts, but it also specifies some dependencies between the four main concepts, namely Topic, Operation, Data and Format. These dependencies are encoded as OWL restrictions. For instance, these dependencies can be seen in the MultiQC tool on Figure 5.2.2, the operation *Nucleic Acid Sequence Analysis* (`operation_2478`) is associated to *Nucleic Acid Sequence* (`data_2977`). As MultiQC is annotated with *Nucleic Acid Sequence Analysis*, we can infer that it takes as input some *Nucleic Acid Sequence* data.

We implemented a specific inference regime for EDAM in the EDAMannot toolbox, allowing to leverage this additional knowledge captured by the EDAM ontology. As an example, for MultiQC, we are now able to enrich the direct annotations stated in bio.tools with inferred transversal EDAM data and format annotation, thus addressing the "Tool adoption", "Workflow input formats" as well as the "Workflow tools output formats" use cases.

5.3. Analysis of annotation informativeness

This analysis aims at computing the informativeness of tools annotations [29] by exploiting all stated and inferred annotations. We grouped tools into three categories based on an arbitrary threshold applied to the sum of their topic and operation entropy ($\max(S_e) = 15.76$). Most tools (29,066; 95.83%) fall into the low or medium information categories. Low-information tools ($S_e < 5$) represent 15,687 entries, while medium-information tools ($5 \leq S_e < 10$) account for 13,379 entries. Only 1,262 tools have an entropy sum above 10.

Annotation quality of tools can be compared using the EDAMannot qc command. For example, in Table 7 we sampled five software tools from different bioinformatics domains to illustrate the variation in annotation informativeness observed in bio.tools.

The tool morFeus[30] has few annotations (around ten when inherited classes are included), and these annotations are not very informative: the sum of the Information Content (IC) for its topics and operations is 7.75, which is far below the median IC value (49.92). SCDO[31] and AFCMHeatMap[32] have a similar number of annotations, but the annotations of AFCMHeatMap are almost twice as informative, when considering IC alone. However, the entropy of SCDO and AFCMHeatMap is similar, because the rarity of AFCMHeatMap annotations is balanced by entropy, which favours more diversified annotations. The software BoMiProt[33] also illustrates that some software entries in bio.tools have a very large number of high-quality annotations, although such tools represent only 1,262 tools (4.16%) of the entire registry. The MultiQC[34] tool shows a relatively low Information Content compared to the SCDO tool. However, it has several diverse annotations, especially for its topics (3.33, compared to the median of 2.78 ± 1.84), and slightly less diverse annotations for its operations (1.57, compared to the median of 1.90 ± 1.32). Overall, the entropy sum for MultiQC (topics and operations) is slightly above the median value of 4.87 ± 2.68. Its total informativeness, considering both Information Content and entropy, remains low. This highlights the need to detect and improve annotation quality, particularly for a widely used tool like MultiQC, which plays a key role in many workflows.

EDAMannot can be particularly useful both: for registry data curators, helping them maintaining a minimal, yet informative, set of tools and workflow annotations; as well as for software agents aiming at searching, re-using, composing, or analyzing scientific workflows.

[29] https://github.com/ulysseLeclanche/EDAMannot/blob/main/Notebooks/Annotation_informativeness_3_categories.ipynb
[30] https://bio.tools/morfeus
[31] https://bio.tools/SCDO
[32] https://bio.tools/AFCMHeatMap
[33] https://bio.tools/bomiprot
[34] https://bio.tools/multiqc

Table 7
Comparison of annotation metrics between different tools sampled in bio.tools; metrics are obtained using the EDAMAnnot QC command.

Tool	Nb. Topics	Nb. Operations	Information content	Entropy
BoMiProt	25	28	197.63	14.15
AFCMHeatMap	9	13	86.01	5.05
SCDO	12	6	49.86	5.10
MultiQC	10	6	33.08	4.90
morFeus	7	3	7.75	3.16

6. Conclusion and future works

This work presents a significant step toward FAIR sharing of scientific workflows, along with their associated metadata and annotations. By aggregating workflows from multiple Workflow Management Systems (WMS), including their execution traces where available, we provide a structured knowledge base that facilitates discovery, understanding, comprehension, reuse, and interoperability of computational workflows. This resource not only enhances reproducibility but also supports researchers in identifying, adapting, and extending existing workflows for their own analyses.

There still remain key limitations, primarily related to the quality and completeness of annotations and metadata. We selected several annotation standards like EFO or SIO as target vocabularies. However, they are not yet used for workflow description in the knowledge base, due to the lack of workflow annotations encoded in those standards in workflow registries. Similarly, while limited by gaps in reference ontologies like EDAM, this work identifies these gaps and contributes to addressing them through targeted curation and maintenance. While our queries yield meaningful insights, particularly for well-annotated Galaxy workflows, the results remain constrained by the depth and accuracy of the available metadata. This challenge is not unique to our project but reflects a broader community-wide issue in workflow and tool annotation. The scientific community increasingly recognises the need for standardised, high-quality annotations, yet achieving this remains an ongoing effort.

Therefore, three main directions can be overtaken in future work: (i) Adding richer, structural metadata and enhancing annotation depth and coverage by integrating a larger diversity of established standards and completing bio.tools annotations, including domain-specific annotations, (ii) Increasing the scale of workflow integration, especially with their execution traces by extending retrospective provenance beyond Galaxy (e.g. to Nextflow, CWL, Snakemake), (iii) Addressing annotation quality, for example by leveraging Large Language Models (LLMs) for semi-automated annotation extraction and standardisation [50], in addition to community-driven curation efforts, e.g., through platforms like WorkflowHub, in collaboration with existing initiatives such as ELIXIR services, WorkflowsCommunity[35] and other community-based efforts.

[35]https://workflows.community

While this study demonstrates the feasibility and value of a centralized, queryable workflow knowledge base, its full potential will only be realized through continued community engagement and improved annotation practices. By fostering collaboration between workflow developers, tool registries, and domain experts, we can move toward a future where scientific workflows are not only reusable but also deeply interconnected and easily discoverable. This will ultimately accelerate reproducible and transparent computational research across the life sciences.

7. Availability and implementation

The datasets and the code used to generate the results are available in the following repositories:

i) https://gitlab.liris.cnrs.fr/sharefair/knowledge_base_workflow_annotations/ShareFAIR-KG (v1.0.0, GNU General Public License v3.0) for extracting workflow metadata, generating data to populate the knowledge base, deploying the knowledge base and executing SPARQL queries on the knowledge base. The exact version used is archived in Software Heritage: https://archive.softwareheritage.org/swh:1:dir:4e761e3e36d350a3a02ea7516ff8813db2d945d6;origin=https://gitlab.liris.cnrs.fr/sharefair/knowledge_base_workflow_annotations/ShareFAIR-KG.git;visit=swh:1:snp:e53f57ef45e971e1b9b76477d1f39f4fbe3de77f;anchor=swh:1:rel:c248bea7d9617f587ef7d2dfb180cb00c4ce7eb2.

ii) https://github.com/ulysseLeclanche/EDAMannot (GNU General Public License v3.0) for the toolbox EDAMannot. The exact version used is archived in Software Heritage: https://archive.softwareheritage.org/swh:1:dir:bf911b652ea16cd37930556b21f6a11e67b82052;origin=https://github.com/ulysseLeclanche/EDAMannot;visit=swh:1:snp:6f6354cb8277bab48068ab9d2aca710390d1b236;anchor=swh:1:rev:1d589fae3c65c9bf3738f542024322220bd466d4.

iii) Knowledge base data content: https://doi.org/10.5281/zenodo.17737888.

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Declaration on Generative AI

During the preparation of this work, the authors used Kimi K2 (Moonshot AI) to: Paraphrase and reword, Improve writing style, Grammar and spelling check. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

[1] S. Cohen-Boulakia, K. Belhajjame, O. Collin, J. Chopard, C. Froidevaux, A. Gaignard, K. Hinsen, P. Larmande, Y. L. Bras, F. Lemoine, F. Mareuil, H. Ménager, C. Pradal, C. Blanchet, Scientific workflows for computational reproducibility in the life sciences: Status, challenges and opportunities, Future Generation Computer Systems 75 (2017) 284–298. doi:10.1016/j.future.2017.01.012.

[2] M. Djaffardjy, G. Marchment, C. Sebe, R. Blanchet, K. Belhajjame, A. Gaignard, F. Lemoine, S. Cohen-Boulakia, Developing and reusing bioinformatics data analysis pipelines using scientific workflow systems, Computational and Structural Biotechnology Journal 21 (2023) 2075–2085. URL: https://www.sciencedirect.com/science/article/pii/S2001037023001010. doi:10.1016/j.csbj.2023.03.003.

[3] P. A. Ewels, A. Peltzer, S. Fillinger, H. Patel, J. Alneberg, A. Wilm, M. U. Garcia, P. Di Tommaso, S. Nahnsen, The nf-core framework for community-curated bioinformatics pipelines, Nature Biotechnology 38 (2020) 276–278. URL: https://www.nature.com/articles/s41587-020-0439-x. doi:10.1038/s41587-020-0439-x, publisher: Nature Publishing Group.

[4] O. J. R. Gustafsson, S. R. Wilkinson, F. Bacall, L. Pireddu, S. Soiland-Reyes, S. Leo, S. Owen, N. Juty, J. M. Fernández, B. Grüning, T. Brown, H. Ménager, S. Capella-Gutierrez, F. Coppens, C. Goble, WorkflowHub: a registry for computational workflows, 2024. URL: http://arxiv.org/abs/2410.06941. doi:10.48550/arXiv.2410.06941, arXiv:2410.06941 [cs].

[5] C. Goble, S. Cohen-Boulakia, S. Soiland-Reyes, D. Garijo, Y. Gil, M. R. Crusoe, K. Peters, D. Schober, FAIR Computational Workflows, Data Intelligence 2 (2020) 108–121. URL: https://doi.org/10.1162/dint_a_00033. doi:10.1162/dint_a_00033.

[6] S. R. Wilkinson, M. Aloqalaa, K. Belhajjame, M. R. Crusoe, B. de Paula Kinoshita, L. Gadelha, D. Garijo, O. J. R. Gustafsson, N. Juty, S. Kanwal, et al., Applying the fair principles to computational workflows, Scientific Data 12 (2025) 328.

[7] M. D. Wilkinson, M. Dumontier, I. J. Aalbersberg, G. Appleton, M. Axton, A. Baak, N. Blomberg, J.-W. Boiten, L. B. da Silva Santos, P. E. Bourne, J. Bouwman, A. J. Brookes, T. Clark, M. Crosas, I. Dillo, O. Dumon, S. Edmunds, C. T. Evelo, R. Finkers, A. Gonzalez-Beltran, A. J. G. Gray, P. Groth, C. Goble, J. S. Grethe, J. Heringa, P. A. C. 't Hoen, R. Hooft, T. Kuhn, R. Kok, J. Kok, S. J. Lusher, M. E. Martone, A. Mons, A. L. Packer, B. Persson, P. Rocca-Serra, M. Roos, R. van Schaik, S.-A. Sansone, E. Schultes, T. Sengstag, T. Slater, G. Strawn, M. A. Swertz, M. Thompson, J. van der Lei, E. van Mulligen, J. Velterop, A. Waagmeester, P. Wittenburg, K. Wolstencroft, J. Zhao, B. Mons, The FAIR Guiding Principles for scientific data management and stewardship, Scientific Data 3 (2016) 160018. doi:10.1038/sdata.2016.18.

[8] M. Barker, N. P. Chue Hong, D. S. Katz, A.-L. Lamprecht, C. Martinez-Ortiz, F. Psomopoulos, J. Harrow, L. J. Castro, M. Gruenpeter, P. A. Martinez, T. Honeyman, Introducing the FAIR Principles for research software, Scientific Data 9 (2022) 622. doi:10.1038/s41597-022-01710-x.

[9] D. Garijo Verdejo, Y. Gil, Augmenting PROV with Plans in P-PLAN: Scientific Processes as Linked Data, in: Proceedings of the 2nd International Workshop on Linked Science |

2nd International Workshop on Linked Science | 12/11/2012 - 12/11/2012 | Boston, USA, volume 951, Facultad de Informática (UPM), Boston, USA, 2012, pp. 0–0. URL: http://ceur-ws.org/Vol-951/.

[10] P. Missier, S. Dey, K. Belhajjame, V. Cuevas-Vicenttin, B. Ludäscher, D-prov, extending the prov provenance model with workflow structure, in: 5th USENIX Workshop on the Theory and Practice of Provenance (TaPP 13), 2013.

[11] A. S. Butt, P. Fitch, ProvONE+: A Provenance Model for Scientific Workflows, in: Z. Huang, W. Beek, H. Wang, R. Zhou, Y. Zhang (Eds.), Web Information Systems Engineering – WISE 2020, Springer International Publishing, 2020, pp. 431–444. doi:10.1007/978-3-030-62008-0_30.

[12] F. Z. Khan, S. Soiland-Reyes, R. O. Sinnott, A. Lonie, C. Goble, M. R. Crusoe, Sharing interoperable workflow provenance: A review of best practices and their practical application in CWLProv, GigaScience 8 (2019) giz095. doi:10.1093/gigascience/giz095.

[13] A. Gaignard, H. Skaf-Molli, K. Belhajjame, Findable and reusable workflow data products: A genomic workflow case study, Semantic Web 11 (2020) 751–763. URL: https://journals.sagepub.com/doi/abs/10.3233/SW-200374. doi:10.3233/SW-200374. arXiv:https://journals.sagepub.com/doi/pdf/10.3233/SW-200374.

[14] S. Leo, M. R. Crusoe, L. Rodríguez-Navas, R. Sirvent, A. Kanitz, P. De Geest, R. Wittner, L. Pireddu, D. Garijo, J. M. Fernández, I. Colonnelli, M. Gallo, T. Ohta, H. Suetake, S. Capella-Gutierrez, R. de Wit, B. P. Kinoshita, S. Soiland-Reyes, Recording provenance of workflow runs with RO-Crate, 2024. doi:10.48550/arXiv.2312.07852. arXiv:2312.07852.

[15] The Galaxy Community, The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2022 update, Nucleic Acids Research 50 (2022) W345–W351. doi:10.1093/nar/gkac247.

[16] P. Di Tommaso, M. Chatzou, E. W. Floden, P. P. Barja, E. Palumbo, C. Notredame, Nextflow enables reproducible computational workflows, Nature Biotechnology 35 (2017) 316–319. URL: https://www.nature.com/articles/nbt.3820. doi:10.1038/nbt.3820, publisher: Nature Publishing Group.

[17] B. Smith, M. Ashburner, C. Rosse, J. Bard, W. Bug, W. Ceusters, L. J. Goldberg, K. Eilbeck, A. Ireland, C. J. Mungall, N. Leontis, P. Rocca-Serra, A. Ruttenberg, S.-A. Sansone, R. H. Scheuermann, N. Shah, P. L. Whetzel, S. Lewis, The OBO Foundry: Coordinated evolution of ontologies to support biomedical data integration, Nature biotechnology 25 (2007) 1251. doi:10.1038/nbt1346.

[18] K. Eilbeck, S. E. Lewis, C. J. Mungall, M. Yandell, L. Stein, R. Durbin, M. Ashburner, The Sequence Ontology: A tool for the unification of genome annotations, Genome Biology 6 (2005) R44. doi:10.1186/gb-2005-6-5-r44.

[19] The Gene Ontology Consortium, S. A. Aleksander, J. Balhoff, S. Carbon, J. M. Cherry, H. J. Drabkin, D. Ebert, M. Feuermann, P. Gaudet, N. L. Harris, D. P. Hill, R. Lee, H. Mi, S. Moxon, C. J. Mungall, A. Muruganugan, T. Mushayahama, P. W. Sternberg, P. D. Thomas, K. Van Auken, J. Ramsey, D. A. Siegele, R. L. Chisholm, P. Fey, M. C. Aspromonte, M. V. Nugnes, F. Quaglia, S. Tosatto, M. Giglio, S. Nadendla, G. Antonazzo, H. Attrill, G. dos Santos, S. Marygold, V. Strelets, C. J. Tabone, J. Thurmond, P. Zhou, S. H. Ahmed, P. Asanitthong, D. Luna Buitrago, M. N. Erdol, M. C. Gage, M. Ali Kadhum, K. Y. C. Li,

M. Long, A. Michalak, A. Pesala, A. Pritazahra, S. C. C. Saverimuttu, R. Su, K. E. Thurlow, R. C. Lovering, C. Logie, S. Oliferenko, J. Blake, K. Christie, L. Corbani, M. E. Dolan, H. J. Drabkin, D. P. Hill, L. Ni, D. Sitnikov, C. Smith, A. Cuzick, J. Seager, L. Cooper, J. Elser, P. Jaiswal, P. Gupta, P. Jaiswal, S. Naithani, M. Lera-Ramirez, K. Rutherford, V. Wood, J. L. De Pons, M. R. Dwinell, G. T. Hayman, M. L. Kaldunski, A. E. Kwitek, S. J. F. Laulederkind, M. A. Tutaj, M. Vedi, S.-J. Wang, P. D'Eustachio, L. Aimo, K. Axelsen, A. Bridge, N. Hyka-Nouspikel, A. Morgat, S. A. Aleksander, J. M. Cherry, S. R. Engel, K. Karra, S. R. Miyasato, R. S. Nash, M. S. Skrzypek, S. Weng, E. D. Wong, E. Bakker, T. Z. Berardini, L. Reiser, A. Auchincloss, K. Axelsen, G. Argoud-Puy, M.-C. Blatter, E. Boutet, L. Breuza, A. Bridge, C. Casals-Casas, E. Coudert, A. Estreicher, M. Livia Famiglietti, M. Feuermann, A. Gos, N. Gruaz-Gumowski, C. Hulo, N. Hyka-Nouspikel, F. Jungo, P. Le Mercier, D. Lieberherr, P. Masson, A. Morgat, I. Pedruzzi, L. Pourcel, S. Poux, C. Rivoire, S. Sundaram, A. Bateman, E. Bowler-Barnett, H. Bye-A-Jee, P. Denny, A. Ignatchenko, R. Ishtiaq, A. Lock, Y. Lussi, M. Magrane, M. J. Martin, S. Orchard, P. Raposo, E. Speretta, N. Tyagi, K. Warner, R. Zaru, A. D. Diehl, R. Lee, J. Chan, S. Diamantakis, D. Raciti, M. Zarowiecki, M. Fisher, C. James-Zorn, V. Ponferrada, A. Zorn, S. Ramachandran, L. Ruzicka, M. Westerfield, The Gene Ontology knowledgebase in 2023, Genetics 224 (2023) iyad031. URL: https://doi.org/10.1093/genetics/iyad031. doi:10.1093/genetics/iyad031.

[20] M. Ashburner, C. A. Ball, J. A. Blake, D. Botstein, H. Butler, J. M. Cherry, A. P. Davis, K. Dolinski, S. S. Dwight, J. T. Eppig, M. A. Harris, D. P. Hill, L. Issel-Tarver, A. Kasarskis, S. Lewis, J. C. Matese, J. E. Richardson, M. Ringwald, G. M. Rubin, G. Sherlock, Gene Ontology: tool for the unification of biology, Nature Genetics 25 (2000) 25–29. URL: https://www.nature.com/articles/ng0500_25. doi:10.1038/75556, publisher: Nature Publishing Group.

[21] S. d. Coronado, M. W. Haber, N. Sioutos, M. S. Tuttle, L. W. Wright, NCI Thesaurus: Using Science-Based Terminology to Integrate Cancer Research Results, in: MEDINFO 2004, IOS Press, 2004, pp. 33–37. URL: https://ebooks.iospress.nl/doi/10.3233/978-1-60750-949-3-33. doi:10.3233/978-1-60750-949-3-33.

[22] K. Degtyarenko, P. de Matos, M. Ennis, J. Hastings, M. Zbinden, A. McNaught, R. Alcántara, M. Darsow, M. Guedj, M. Ashburner, ChEBI: a database and ontology for chemical entities of biological interest, Nucleic Acids Research 36 (2008) D344–D350. URL: https://doi.org/10.1093/nar/gkm791. doi:10.1093/nar/gkm791.

[23] M. Duering, G. J. Biessels, A. Brodtmann, C. Chen, C. Cordonnier, F.-E. de Leeuw, S. Debette, R. Frayne, E. Jouvent, N. S. Rost, A. ter Telgte, R. Al-Shahi Salman, W. H. Backes, H.-J. Bae, R. Brown, H. Chabriat, A. De Luca, C. deCarli, A. Dewenter, F. N. Doubal, M. Ewers, T. S. Field, A. Ganesh, S. Greenberg, K. G. Helmer, S. Hilal, A. C. C. Jochems, H. Jokinen, H. Kuijf, B. Y. K. Lam, J. Lebenberg, B. J. MacIntosh, P. Maillard, V. C. T. Mok, L. Pantoni, S. Rudilosso, C. L. Satizabal, M. D. Schirmer, R. Schmidt, C. Smith, J. Staals, M. J. Thrippleton, S. J. van Veluw, P. Vemuri, Y. Wang, D. Werring, M. Zedde, R. O. Akinyemi, O. H. Del Brutto, H. S. Markus, Y.-C. Zhu, E. E. Smith, M. Dichgans, J. M. Wardlaw, Neuroimaging standards for research into small vessel disease—advances since 2013, The Lancet Neurology 22 (2023) 602–618. URL: https://www.sciencedirect.com/science/article/pii/S147444222300131X. doi:10.1016/S1474-4422(23)00131-X.

[24] J. M. Wardlaw, E. E. Smith, G. J. Biessels, C. Cordonnier, F. Fazekas, R. Frayne, R. I. Lindley, J. T. O'Brien, F. Barkhof, O. R. Benavente, S. E. Black, C. Brayne, M. Breteler, H. Chabriat, C. DeCarli, F.-E. de Leeuw, F. Doubal, M. Duering, N. C. Fox, S. Greenberg, V. Hachinski, I. Kilimann, V. Mok, R. v. Oostenbrugge, L. Pantoni, O. Speck, B. C. M. Stephan, S. Teipel, A. Viswanathan, D. Werring, C. Chen, C. Smith, M. van Buchem, B. Norrving, P. B. Gorelick, M. Dichgans, Neuroimaging standards for research into small vessel disease and its contribution to ageing and neurodegeneration, The Lancet Neurology 12 (2013) 822–838. URL: https://www.sciencedirect.com/science/article/pii/S1474442213701248. doi:10.1016/S1474-4422(13)70124-8.

[25] A. Barton, A. Rosier, A. Burgun, J.-F. Ethier, ois, The Cardiovascular Disease Ontology, in: Formal Ontology in Information Systems, IOS Press, 2014, pp. 409–414. URL: https://ebooks.iospress.nl/doi/10.3233/978-1-61499-438-1-409. doi:10.3233/978-1-61499-438-1-409.

[26] W. J. Bug, G. A. Ascoli, J. S. Grethe, A. Gupta, C. Fennema-Notestine, A. R. Laird, S. D. Larson, D. Rubin, G. M. Shepherd, J. A. Turner, M. E. Martone, The NIFSTD and BIRNLex Vocabularies: Building Comprehensive Ontologies for Neuroscience, Neuroinformatics 6 (2008) 175–194. doi:10.1007/s12021-008-9032-z.

[27] The Bioconda Team, B. Grüning, R. Dale, A. Sjödin, B. A. Chapman, J. Rowe, C. H. Tomkins-Tinch, R. Valieris, J. Köster, Bioconda: sustainable and comprehensive software distribution for the life sciences, Nature Methods 15 (2018) 475–476. URL: https://www.nature.com/articles/s41592-018-0046-7. doi:10.1038/s41592-018-0046-7.

[28] J. Bai, C. Bandla, J. Guo, R. Vera Alvarez, M. Bai, J. A. Vizcaíno, P. Moreno, B. Grüning, O. Sallou, Y. Perez-Riverol, Biocontainers registry: Searching bioinformatics and proteomics tools, packages, and containers, Journal of Proteome Research 20 (2021) 2056–2061. URL: https://doi.org/10.1021/acs.jproteome.0c00904. doi:10.1021/acs.jproteome.0c00904. arXiv:https://doi.org/10.1021/acs.jproteome.0c00904, pMID: 33625229.

[29] S. Möller, H. N. Krabbenhöft, A. Tille, D. Paleino, A. Williams, K. Wolstencroft, C. Goble, R. Holland, D. Belhachemi, C. Plessy, Community-driven computational biology with Debian Linux, BMC bioinformatics 11 Suppl 12 (2010) S5. doi:10.1186/1471-2105-11-S12-S5.

[30] K. David, H. Karl, G. Satrajit, A. Tibor, M. Camille, D. Samir, F. Guillaume, N. Thomas, G. Krzysztof, T. Jessica, K. David, P. Jean-Baptiste, N. B, Describing Assessments and Experiment Metadata with the Neuroimaging Data Model (NIDM), Frontiers in Neuroinformatics 10 (2016). URL: http://www.frontiersin.org/Community/AbstractDetails.aspx?ABS_DOI=10.3389/conf.fninf.2016.20.00069. doi:10.3389/conf.fninf.2016.20.00069.

[31] J. Malone, A. Brown, A. L. Lister, J. Ison, D. Hull, H. Parkinson, R. Stevens, The Software Ontology (SWO): a resource for reproducibility in biomedical data analysis, curation and digital preservation, Journal of Biomedical Semantics 5 (2014) 25. URL: https://doi.org/10.1186/2041-1480-5-25. doi:10.1186/2041-1480-5-25.

[32] K. J. Gorgolewski, T. Auer, V. D. Calhoun, R. C. Craddock, S. Das, E. P. Duff, G. Flandin, S. S. Ghosh, T. Glatard, Y. O. Halchenko, D. A. Handwerker, M. Hanke, D. Keator, X. Li, Z. Michael, C. Maumet, B. N. Nichols, T. E. Nichols, J. Pellman, J.-B. Poline, A. Rokem, G. Schaefer, V. Sochat, W. Triplett, J. A. Turner, G. Varoquaux, R. A. Poldrack, The brain

imaging data structure, a format for organizing and describing outputs of neuroimaging experiments, Scientific Data 3 (2016) 160044. doi:10.1038/sdata.2016.44.

[33] O. Rübel, A. Tritt, R. Ly, B. K. Dichter, S. Ghosh, L. Niu, P. Baker, I. Soltesz, L. Ng, K. Svoboda, L. Frank, K. E. Bouchard, The Neurodata Without Borders ecosystem for neurophysiological data science, eLife 11 (2022) e78362. URL: https://doi.org/10.7554/eLife.78362. doi:10.7554/eLife.78362, publisher: eLife Sciences Publications, Ltd.

[34] J. Ison, M. Kalaš, I. Jonassen, D. Bolser, M. Uludag, H. McWilliam, J. Malone, R. Lopez, S. Pettifer, P. Rice, EDAM: An ontology of bioinformatics operations, types of data and identifiers, topics and formats, Bioinformatics 29 (2013) 1325–1332. doi:10.1093/bioinformatics/btt113.

[35] M. Black, L. Lamothe, H. Eldakroury, M. Kierkegaard, A. Priya, A. Machinda, U. S. Khanduja, D. Patoliya, R. Rathi, T. P. C. Nico, G. Umutesi, C. Blankenburg, A. Op, P. Chieke, O. Babatunde, S. Laurie, S. Neumann, V. Schwammle, I. Kuzmin, C. Hunter, J. Karr, J. Ison, A. Gaignard, B. Brancotte, H. Menager, M. Kalas, Edam: The bioscientific data analysis ontology (update 2021), F1000Research 11 (2021). doi:10.7490/f1000research.1118900.1.

[36] J. Ison, K. Rapacki, H. Ménager, M. Kalaš, E. Rydza, P. Chmura, C. Anthon, N. Beard, K. Berka, D. Bolser, T. Booth, A. Bretaudeau, J. Brezovsky, R. Casadio, G. Cesareni, F. Coppens, M. Cornell, G. Cuccuru, K. Davidsen, G. D. Vedova, T. Dogan, O. Doppelt-Azeroual, L. Emery, E. Gasteiger, T. Gatter, T. Goldberg, M. Grosjean, B. Grüning, M. Helmer-Citterich, H. Ienasescu, V. Ioannidis, M. C. Jespersen, R. Jimenez, N. Juty, P. Juvan, M. Koch, C. Laibe, J.-W. Li, L. Licata, F. Mareuil, I. Mičetić, R. M. Friborg, S. Moretti, C. Morris, S. Möller, A. Nenadic, H. Peterson, G. Profiti, P. Rice, P. Romano, P. Roncaglia, R. Saidi, A. Schafferhans, V. Schwämmle, C. Smith, M. M. Sperotto, H. Stockinger, R. S. Vařeková, S. C. Tosatto, V. de la Torre, P. Uva, A. Via, G. Yachdav, F. Zambelli, G. Vriend, B. Rost, H. Parkinson, P. Løngreen, S. Brunak, Tools and data services registry: a community effort to document bioinformatics resources, Nucleic Acids Research 44 (2016) D38–D47. URL: https://doi.org/10.1093/nar/gkv1116. doi:10.1093/nar/gkv1116.

[37] J. Ison, H. Ienasescu, P. Chmura, E. Rydza, H. Ménager, M. Kalaš, V. Schwämmle, B. Grüning, N. Beard, R. Lopez, S. Duvaud, H. Stockinger, B. Persson, R. S. Vařeková, T. Raček, J. Vondrášek, H. Peterson, A. Salumets, I. Jonassen, R. Hooft, T. Nyrönen, A. Valencia, S. Capella, J. Gelpí, F. Zambelli, B. Savakis, B. Leskošek, K. Rapacki, C. Blanchet, R. Jimenez, A. Oliveira, G. Vriend, O. Collin, J. van Helden, P. Løngreen, S. Brunak, The bio.tools registry of software tools and data resources for the life sciences, Genome Biology 20 (2019) 164. URL: https://doi.org/10.1186/s13059-019-1772-6. doi:10.1186/s13059-019-1772-6.

[38] G. Marchment, M. Schmit, C. Sebe, F. Lemoine, H. Ménager, S. Cohen-Boulakia, Representing bioinformatics Nextflow workflows in RO-Crate : Challenges and opportunities, 2024. doi:10.5281/zenodo.10822156.

[39] J. Malone, E. Holloway, T. Adamusiak, M. Kapushesky, J. Zheng, N. Kolesnikov, A. Zhukova, A. Brazma, H. Parkinson, Modeling sample variables with an Experimental Factor Ontology, Bioinformatics 26 (2010) 1112–1118. doi:10.1093/bioinformatics/btq099.

[40] M. Dumontier, C. J. Baker, J. Baran, A. Callahan, L. Chepelev, J. Cruz-Toledo, N. R. Del Rio, G. Duck, L. I. Furlong, N. Keath, D. Klassen, J. P. McCusker, N. Queralt-Rosinach, M. Samwald, N. Villanueva-Rosales, M. D. Wilkinson, R. Hoehndorf, The Semanticscience Integrated Ontology (SIO) for biomedical research and knowledge discovery, Journal of Biomedical Semantics 5 (2014) 14. doi:10.1186/2041-1480-5-14.

[41] P. Amstutz, M. R. Crusoe, N. Tijanić, B. Chapman, J. Chilton, M. Heuer, A. Kartashov, D. Leehr, H. Ménager, M. Nedeljkovich, M. Scales, S. Soiland-Reyes, L. Stojanovic, Common Workflow Language, v1.0, 2016. URL: https://research.manchester.ac.uk/en/publications/common-workflow-language-v10/. doi:10.6084/m9.figshare.3115156.v2, publisher: figshare.

[42] M. R. Crusoe, S. Abeln, A. Iosup, P. Amstutz, J. Chilton, N. Tijanić, H. Ménager, S. Soiland-Reyes, B. Gavrilović, C. Goble, et al., Methods included: standardizing computational reuse and portability with the common workflow language, Communications of the ACM 65 (2022) 54–63.

[43] S. Bray, J. Chilton, M. Bernt, N. Soranzo, M. v. d. Beek, B. Batut, H. Rasche, M. Čech, P. J. A. Cock, B. Grüning, A. Nekrutenko, The Planemo toolkit for developing, deploying, and executing scientific data analyses in Galaxy and beyond, Genome Research 33 (2023) 261–268. URL: http://genome.cshlp.org/content/33/2/261. doi:10.1101/gr.276963.122, company: Cold Spring Harbor Laboratory Press Distributor: Cold Spring Harbor Laboratory Press Institution: Cold Spring Harbor Laboratory Press Label: Cold Spring Harbor Laboratory Press Publisher: Cold Spring Harbor Lab.

[44] G. Marchment, B. Brancotte, M. Schmit, F. Lemoine, S. Cohen-Boulakia, Bioflow-insight: facilitating reuse of nextflow workflows with structure reconstruction and visualization, NAR Genomics and Bioinformatics 6 (2024) lqae092. URL: https://doi.org/10.1093/nargab/lqae092. doi:10.1093/nargab/lqae092.

[45] H.-I. Ienasescu, S. Capella-Gutierrez, F. Coppens, J. M. Fernandez, A. Gaignard, C. Goble, B. Grüning, O. J. R. Gustafsson, J. L. Gelpí, J. Harrow, et al., The ELIXIR Research Software Ecosystem, in: 24th annual Bioinformatics Open Source Conference (BOSC 2023), 2023. URL: https://doi.org/10.7490/f1000research.1119585.1. doi:10.7490/f1000research.1119585.1.

[46] P. Ewels, M. Magnusson, S. Lundin, M. Käller, Multiqc: summarize analysis results for multiple tools and samples in a single report, Bioinformatics 32 (2016) 3047–3048. URL: https://doi.org/10.1093/bioinformatics/btw354. doi:10.1093/bioinformatics/btw354.

[47] B. J. Callahan, P. J. McMurdie, M. J. Rosen, A. W. Han, A. J. A. Johnson, S. P. Holmes, Dada2: High-resolution sample inference from illumina amplicon data, Nature methods 13 (2016) 581–583.

[48] L. Lopez-Delisle, github.com/iwc-workflows/atacseq/main, 2025. URL: https://doi.org/10.5281/zenodo.15078780. doi:10.5281/zenodo.15078780.

[49] U. Le Clanche, S. Cohen-Boulakia, Y. Le Cunff, O. Dameron, A. Gaignard, Assessing bioinformatics software annotations: bio. tools case-study, in: JOBIM 2025-Journées Ouvertes en Biologie, Informatique et Mathématiques, 2025, pp. 194–200.

[50] E. M. del Pico, J. L. Gelpí, S. Capella-Gutiérrez, Identity resolution of software metadata using large language models, 2025. URL: https://arxiv.org/abs/2505.23500.

arXiv:2505.23500.